

ADA-Marker Answer Sheet

Anonymized local demo

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PROBLEM

Q2

Answer

3. We can build a reversed-Z-value algorithm. This time, $\text{Z}[i] = j - \text{Z}(j)$, $1 \leq j \leq n$
 $\text{Z}[i] = j$ that satisfy $\text{Z}[i]$. By implementing the ~~same~~ ^{similar} solution in Z-value
algorithm, the three cases can also be solved in reversed-Z-value
algorithm. And, thus, the time complexity will be $O(n)$. However,
this time we not only compare the string from the head, but
we also compare strings started at the second, third... character.
So that we can deal with cases that the longest palindrome is
not started at the very beginning. And, thus, the overall time
complexity will be $O(n) \cdot O(n) = O(n^2)$